Paulo Moura Martins



LLM Engineer · AI Agent Developer · Senior Software Engineer · Data Engineer

Montreal, Quebec, Canada · Canadian Permanent Resident · Open to remote (North America / EMEA overlap)
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Languages: Portuguese (Native) · English (Advanced, C1) · French (Advanced, C1) · Spanish (Advanced, C1)

PROFESSIONAL SUMMARY

Senior engineer with 15+ years building and operating production software, now focused full-time on LLM systems: agent orchestration, retrieval-augmented generation, evaluation infrastructure, and the GPU-side engineering that makes model serving economical. Spent a decade on high-throughput ERP platforms in Brazil (Sankhya, TOTVS) before relocating to Canada in 2021 to join Shopify, where I worked across merchant-facing product, platform reliability, and the internal ML/LLM tooling that shipped generative commerce features to millions of stores. Since 2025 I have worked as an independent contractor for frontier AI labs and their research partners, designing evaluation rubrics, RL environments, and adversarial datasets for code, ML-systems, and security domains. Comfortable owning a problem end-to-end: framing it, instrumenting it, shipping it, and proving it moved a number.

CORE SKILLS

LLM & Agent Engineering: Agent orchestration & multi-agent planning · Function / tool calling · Model Context Protocol (MCP) · RAG architectures (hybrid BM25 + dense, re-ranking, chunking strategy) · Prompt engineering & systematic prompt evaluation · Structured output & constrained decoding · Guardrails, refusal calibration, jailbreak resistance · Long-context & context-window budgeting

ML Systems & Inference: PyTorch · JAX · Hugging Face Transformers · vLLM · SGLang · TensorRT-LLM · Ray Serve · KV-cache & paged attention · Continuous batching · Quantization (GPTQ, AWQ, FP8) · Custom GPU kernels (CUDA, Triton) · torch.profiler / Nsight / Kineto trace analysis · Distributed training (FSDP, DDP, DeepSpeed) · A100 / H100 / L4 / TPU v4

Evaluation & Research Ops: Eval harness design · Rubric & annotation-schema authoring · LLM-as-judge calibration · Side-by-side (SxS) preference ranking · Inter-annotator agreement & signal validation · Failure-mode taxonomy & root-cause analysis · RL environments & reward shaping · RLHF / DPO data pipelines · Benchmark construction (SWE-bench-style, agentic tasks)

Languages: Python · TypeScript · JavaScript · Node.js · C# / .NET · Ruby · Go · Java · SQL · PL/SQL · Dart (Flutter) · Bash

Data Engineering: dbt · Apache Kafka · Apache Spark · Airflow · Snowflake · BigQuery · Redshift · Vitess / MySQL · PostgreSQL · Oracle · Redis · Pinecone / pgvector / Weaviate · CDC & streaming ETL · Dimensional modelling

Platform & Security: GCP · AWS · Azure · Kubernetes · Docker & Docker Compose · Terraform · CI/CD (GitHub Actions, Buildkite) · Observability (Datadog, Prometheus, Grafana, OpenTelemetry) · Threat modelling · SAST/DAST · CVE reproduction & remediation · OWASP Top 10 · CVSS / CWE / CAPEC

Product & Domain: WhatsApp Business API · Stripe & payments · Multi-tenant SaaS · ERP & fiscal compliance (Brazil) · E-commerce at scale · Digital health product development · Technical writing & stakeholder communication

PROFESSIONAL EXPERIENCE

Independent Contractor , Frontier AI Labs & Research Partners

Remote (Montreal, QC) · Feb 2025 – Present

Contract technical staff for frontier-model training and evaluation programs. Engagements span code evaluation, ML-systems task authoring, security dataset QA, and personalization-quality analysis.

- Authored and reviewed 1,400+ software-engineering evaluation tasks across Python, TypeScript/React, Go, Rust, C++ and Java for a frontier lab's code-capability program; my task set became a standing calibration pack used to onboard 60+ downstream contributors after it produced the highest reviewer-acceptance rate on the project (96.2% vs. a 78% program average).
- Designed a two-part verification framework for agentic coding tasks , a functional test suite plus an adversarial regression suite , that cut false-positive "solved" signals by 41% on an internal benchmark of 900 tasks, removing a known source of reward hacking in the training loop.

- Built ML-systems evaluation content for an MLOps program covering GPU kernel optimization, profiler-trace interpretation, distributed-training debugging, and inference-serving trade-offs; wrote reference solutions in CUDA and Triton and rubrics that graded reasoning about throughput, latency and HBM-bandwidth limits rather than final answers alone.
- Led QA on a vulnerability-reproduction dataset: assessed 300+ CVE reproductions for fidelity, verified that Docker Compose lab environments genuinely recreated exploitable conditions, and rejected or repaired 22% of submissions where the "fix" passed tests without closing the vulnerability class (SSRF, insecure deserialization, command injection, privilege escalation).
- Ran side-by-side personalization evaluations for a consumer assistant feature, designing multi-turn prompts grounded in real personal context and writing structured rationales on grounding failures, forced inferences and over-narration; findings fed a rubric revision that lifted inter-annotator agreement from 0.61 to 0.84 Krippendorff's alpha.
- Standing quality gate on two programs: empowered to pause delivery when signal integrity was insufficient, and did so twice , once over a contaminated eval split, once over a rubric that rewarded verbosity over correctness.

Shopify Inc.

Montreal, QC, Canada · Senior Software Engineer → Staff Software Engineer, Commerce AI · Mar 2021 – Jan 2025

Global commerce platform serving millions of merchants. Relocated from Brazil to join the Montreal office; moved from core merchant product into the ML/LLM platform group in 2023.

- Tech lead for the retrieval and tool-calling layer behind a merchant-facing AI assistant: designed a hybrid RAG pipeline (BM25 + dense retrieval over pgvector, cross-encoder re-ranking) across help-centre content, merchant catalogue data and admin APIs, reaching 89% answer-groundedness on a 2,000-question internal eval and cutting escalations to human support by 31% in the pilot cohort.
- Built the function-calling contract layer that let the assistant safely mutate merchant state , bulk price edits, inventory adjustments, discount creation , with typed schemas, dry-run previews and a reversible action log; zero P1 incidents attributable to model-initiated writes across 18 months and ~4M executed actions.
- Cut assistant inference cost 63% ($2.1M annualized) by moving the serving tier to vLLM with continuous batching and paged attention, adding prefix-cache reuse for the shared system prompt, and routing 70% of traffic to a distilled small model behind a confidence-gated fallback.
- Profiled and optimized the embedding backfill that indexed 400M+ product records: replaced a naive per-row path with a batched pipeline, wrote a fused Triton kernel for the normalization + pooling step, and reduced a 30-hour nightly job to 4.5 hours on the same H100 allocation.
- Owned the offline evaluation harness for generative commerce features , golden datasets, LLM-as-judge with human-calibrated agreement checks, regression gates wired into CI , which became the required pre-ship gate for four product teams and caught 11 quality regressions before release.
- Earlier tenure (2021–2023): shipped merchant-admin features in Ruby on Rails, React/TypeScript and GraphQL, and React Native surfaces in the Shopify mobile app; led a Vitess shard-migration workstream for a high-write subsystem that removed a projected capacity ceiling 14 months out.
- Mentored six engineers, three to promotion; ran the group's incident-review practice and wrote the internal guide on evaluating LLM features that is still referenced in onboarding.

TOTVS S.A.

São Paulo / Remote, Brazil · Software Engineer → Technical Lead, Platform & Integrations · Jun 2015 – Feb 2021

Largest enterprise software company in Latin America (ERP: Protheus, RM, Datasul; platform: Fluig, TOTVS Cloud), serving 40,000+ corporate clients.

- Led the integration platform team that connected Protheus and RM ERP instances to third-party systems, delivering a Node.js/TypeScript API gateway and event bus (Kafka) that replaced 200+ bespoke point-to-point integrations and reduced average client onboarding from 11 weeks to 3.
- Architected a fiscal-document processing service handling NF-e/NFS-e issuance and SEFAZ reconciliation at 6M+ documents/month, with idempotent retries and a dead-letter workflow that took integration-related support tickets down 47% year over year.
- Built the company's first customer-facing analytics pipeline on Spark, Airflow and Redshift , CDC from Oracle and SQL Server into a dimensional warehouse, dbt for transformations , surfacing operational dashboards to 3,000+ client organizations.
- Modernized a legacy C#/.NET billing module into containerized services on Kubernetes with blue-green deploys; release frequency moved from quarterly to weekly and rollback time from hours to under 4 minutes.

- Ran the internal security-hardening program after a penetration-test finding: introduced SAST/DAST gating in CI, threat-modelling sessions per epic, and a remediation SLA; critical-severity findings in production dropped from 23 to 2 across four quarters.
- Managed a team of nine across three Brazilian sites, set the technical roadmap for the integrations domain, and represented engineering in quarterly planning with product and regional sales leadership.

Sankhya Gestão de Negócios

Uberlândia, MG, Brazil · Junior Developer → Senior Software Engineer / DevOps · Jan 2010 – May 2015

Brazilian ERP vendor headquartered in Uberlândia, building the Sankhya Om management suite for mid-market and enterprise clients across manufacturing, distribution and retail.

- Full-stack development on the core ERP: Java back-end services, Oracle PL/SQL stored procedures and reporting, and the JavaScript front-end for finance, inventory and procurement modules used daily by 15,000+ end users.
- Rewrote the inventory-costing engine (weighted average and FIFO) after profiling exposed a quadratic query path; month-end close for the largest client went from 9 hours to 22 minutes and the fix shipped to the whole install base.
- Introduced the company's first CI pipeline and automated regression suite (Jenkins, 1,800 tests), taking a manual two-day release ritual to a scripted 40-minute deploy and cutting post-release hotfixes by roughly half.
- Built the deployment and monitoring tooling for on-premise and hosted client environments , provisioning scripts, backup verification, Nagios/Zabbix alerting , and carried the on-call pager for tier-one accounts.
- Acted as escalation engineer for critical client incidents, travelling on-site for manufacturing and distribution customers; wrote the internal runbook series still used for Oracle performance triage.

SELECTED PERSONAL & OPEN-SOURCE PROJECTS

Atendo , WhatsApp AI Agent Platform (2023 – Present)

Side product: multi-tenant agent platform on the WhatsApp Business API with function calling into client CRMs, booking systems and payment links. Node.js/TypeScript, Redis-backed conversation state, per-tenant RAG over client knowledge bases, human-handoff queue. ~40 small businesses in Brazil and Quebec, ~120k conversations handled, 74% fully automated resolution.

evalkit , Open-Source LLM Evaluation Harness (2024 – Present)

Python library for building reproducible eval suites: dataset versioning, judge calibration against human labels, cost/latency tracking per run, and CI-friendly regression gates. 1.6k GitHub stars, used by three commercial teams.

kernel-notes , Triton & CUDA Study Repo (2024 – Present)

Annotated implementations of fused attention, layer-norm and quantized matmul kernels with Nsight traces and roofline analysis for each, written as teaching material for engineers moving from application code into GPU work.

EDUCATION

Master of Science in Information Technology (MSIT)

University of the People , Pasadena, California, USA · 2018 – 2020

Focus: distributed systems, data management and software architecture. Capstone: streaming ETL for multi-tenant ERP telemetry.

Bachelor of Science in Computer Science

Universidade Federal de Uberlândia (UFU) , Uberlândia, MG, Brazil · 2007 – 2010

Undergraduate research in database query optimization; teaching assistant for Algorithms and Data Structures.

CERTIFICATIONS

- Google Cloud Professional Machine Learning Engineer , Google Cloud (2024)
- AWS Certified Solutions Architect – Professional , Amazon Web Services (2023)
- Microsoft Certified: Azure Solutions Architect Expert , Microsoft (2022)
- AWS Certified DevOps Engineer – Professional , Amazon Web Services (2022)
- Certified Kubernetes Administrator (CKA) , The Linux Foundation / CNCF (2021)
- Offensive Security Certified Professional (OSCP) , OffSec (2021)
- MCSD: App Builder , C# / .NET , Microsoft (2017)
- Oracle Database SQL Certified Associate (1Z0-071) , Oracle (2014)

PROFESSIONAL DEVELOPMENT

- Digital Health Product Development (SOM-XCHE0025) , Stanford School of Medicine, Stanford Medicine Health Education (2024). 10 AMA PRA Category 1 Credits™; needs finding, regulatory pathways (FDA / CDS exclusions), MVP planning and clinical-trial design for digital health products.
- Neural Networks: Zero to Hero / GPU Mode kernel track , self-directed, 2023–2024.
- Certified Scrum Product Owner (CSPO) , Scrum Alliance, 2019.

ADDITIONAL INFORMATION

Speaking & Writing: Speaker, QCon São Paulo 2020 , "Killing 200 Point-to-Point Integrations"; Shopify internal tech-talk series on LLM evaluation (2024); occasional technical essays at paulomoura.dev.

Open Source: Maintainer of evalkit; contributor to vLLM (docs and two serving-path fixes) and to dbt community packages.

Availability: Full-time, 40 h/week, flexible across PST–EST overlap. Authorized to work in Canada without sponsorship.

References available on request.